Gradient-Based Optimisation for Variational Inference: A Comparative Study Across Linear, Quadratic, and Logistic Regression

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Abstract

This report applies the gradient-based optimisation methods from ENEL445 to variational inference across three Bayesian regression settings: linear, quadratic, and logistic regression. The unifying objective is maximising the Evidence Lower Bound (ELBO), which converts intractable posterior inference into a tractable optimisation problem.

Four optimisation methods are compared in each setting: Coordinate Ascent Variational Inference (CAVI), gradient ascent with Armijo backtracking, Newton's method with regularised finite-difference Hessian, and BFGS with Wolfe conditions. The variational family is $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with a Cholesky reparameterisation to ensure positive definiteness throughout optimisation.

Reference posteriors are obtained from Gibbs samplers. For the linear and quadratic cases, the conjugate structure admits closed-form Gibbs updates; the variational family is a Normal–Gamma approximation. For the logistic case, the Jaakkola–Jordan (Jaakkola & Jordan, 2000) variational bound restores tractability, and a Pólya–Gamma Gibbs sampler (Polson et al., 2013) provides the reference.

The three test cases form a designed progression in inferential difficulty: the linear case has a conjugate Gaussian likelihood ($n = 50$, $p = 2$), the quadratic case adds nonlinearity via feature expansion ($n = 100$, $p = 3$), and the logistic case introduces a non-conjugate Bernoulli likelihood ($n = 200$, $p = 2$). Posterior variance collapse is mild in the linear case, moderate in the quadratic case, and minimal in the logistic case. CAVI is the most efficient method across all three settings. BFGS is the recommended gradient-based alternative when closed-form CAVI updates are not available.

Contents

Introduction

Bayesian inference provides principled uncertainty quantification for regression models, but requires computing the posterior distribution

$$p(\boldsymbol{\beta} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{\beta})\, p(\boldsymbol{\beta}),$$

which is analytically tractable only for conjugate likelihood–prior pairs. For most real-world models the posterior integral

$$p(\boldsymbol{\beta} \mid \boldsymbol{y}) = \frac{p(\boldsymbol{y} \mid \boldsymbol{\beta})\, p(\boldsymbol{\beta})}{\displaystyle\int p(\boldsymbol{y} \mid \boldsymbol{\beta})\, p(\boldsymbol{\beta})\, \mathrm{d}\boldsymbol{\beta}}$$

cannot be evaluated analytically. Variational Inference (VI) converts posterior computation into optimisation by finding the best approximation $q(\boldsymbol{\beta})$ from a tractable family, maximising the Evidence Lower Bound (ELBO):

$$\mathcal{L}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{\beta})] - \mathrm{KL}(q\|p) \leq \log p(\boldsymbol{y}).$$

This project applies the optimisation methods taught in ENEL445 to three Bayesian regression settings, forming a deliberate progression in inferential difficulty.

Case Study 1 — Linear Regression: Gaussian likelihood with conjugate prior; ELBO has a closed-form gradient. Posterior variance collapse is the primary concern. Reference posterior from a conjugate Gibbs sampler. ($n = 50$, $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)$.)

Case Study 2 — Quadratic Regression: Identical Bayesian framework to the linear case, but with feature expansion $\boldsymbol{x} \mapsto (1, x, x^2)$ so the model remains linear in the coefficient vector. Moderate posterior variance collapse. ($n = 100$, $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0, -1.0)$.)

Case Study 3 — Logistic Regression: Non-conjugate Bernoulli likelihood; the Jaakkola–Jordan bound provides a tractable ELBO. Pólya–Gamma Gibbs sampler as the reference. Posterior variance collapse is minimal. ($n = 200$, $\boldsymbol{\beta}_{\text{true}} = (-1.0, 2.0)$.)

All three studies share the same four optimisation methods (CAVI, gradient ascent, Newton, BFGS), the same Cholesky reparameterisation for positive-definiteness, and the same convergence criteria. The unified methodology allows cross-case comparisons of convergence speed, computational cost, posterior accuracy, and variance collapse behaviour.
A known failure mode of mean-field VI is *posterior variance collapse*: the optimised approximation is systematically overconfident, underestimating posterior uncertainty. This failure mode is observed and analysed across all three settings.

1 Common Methodology

1.1 Variational Inference as Optimisation

The mean-field variational approximation factorises the joint posterior as $q(\boldsymbol{\beta}, \tau_e) = q(\boldsymbol{\beta})\, q(\tau_e)$ for the Gaussian regression cases, or $q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ for the logistic case. Maximising the ELBO over the variational parameters $\boldsymbol{\phi}$ is equivalent to minimising the KL divergence $\mathrm{KL}(q\|p)$ between the approximation and the true posterior.

1.2 Constraint Handling: Cholesky Reparameterisation

The covariance matrix $\boldsymbol{\Sigma}$ must remain symmetric positive definite at every iteration. The set of symmetric positive definite matrices has a curved boundary: a standard Euclidean gradient step can cross outside it, producing an invalid covariance. The solution is to parameterise over the Cholesky factor $\boldsymbol{L}$, where $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$ and $\boldsymbol{L}$ is lower-triangular with positive diagonal entries. The diagonal entries are reparameterised as $L_{ii} = e^{\tilde{\ell}_{ii}}$ to allow unconstrained optimisation. This converts the constraint into a smooth bijection and ensures every gradient step produces a valid covariance.

1.3 Optimisation Methods

All four methods share the same objective (the ELBO), the same reparameterisation, and the same convergence criteria.

CAVI: Coordinate Ascent Variational Inference. Closed-form coordinate updates derived from the exponential family structure. Each update exactly maximises the ELBO over one block of parameters while holding the others fixed. Most efficient method; requires conjugate structure.

Gradient ascent: First-order method. Search direction is the gradient $\boldsymbol{d}^{(t)} = \nabla\mathcal{L}(\boldsymbol{\phi}^{(t)})$. Step size determined by Armijo backtracking (sufficient increase condition).

Newton's method: Second-order method. Search direction is $\boldsymbol{d}^{(t)} = -\boldsymbol{H}^{-1}\nabla\mathcal{L}$, where $\boldsymbol{H}$ is the Hessian approximated by finite differences (25 evaluations per iteration). Fast convergence near the optimum; expensive per iteration.

BFGS: Quasi-Newton method. Builds a curvature approximation from gradient differences, requiring only gradient evaluations. Step size determined by Wolfe conditions to maintain positive definiteness of the approximate inverse Hessian ($\boldsymbol{y}_t^T\boldsymbol{s}_t > 0$). Best practical compromise between convergence speed and cost.

For all line-search methods, the search direction must be an ascent direction: $\nabla\mathcal{L}(\boldsymbol{\phi}^{(t)})^T\boldsymbol{d}^{(t)} > 0$.

1.4 Convergence Criteria

Iteration terminates when any of the following conditions is satisfied:

(i) Gradient norm: $\|\nabla\mathcal{L}\| < \varepsilon_g$.

(ii) Relative ELBO change: $|\Delta\mathcal{L}|/|\mathcal{L}| < \varepsilon_r$.

(iii) Parameter change: $\|\Delta\boldsymbol{\phi}\| < \varepsilon_\phi$.

(iv) Maximum iterations: $t \geq t_{\max}$.

CAVI additionally monitors the maximum relative change in variational parameters.

1.5 Reference Samplers

Two types of Gibbs sampler are used as gold-standard references.

Conjugate Gibbs (linear and quadratic cases). The Normal–Gamma conjugate structure of the Bayesian regression model (Gaussian likelihood with Gaussian prior on $\boldsymbol{\beta}$ and Gamma prior on precision τ_e) admits closed-form Gibbs updates:

$$\boldsymbol{\beta} \mid \tau_e, \boldsymbol{y} \sim \mathcal{N}(\boldsymbol{\mu}_n, \boldsymbol{\Sigma}_n), \tag{1}$$

$$\tau_e \mid \boldsymbol{\beta}, \boldsymbol{y} \sim \mathrm{Gamma}(a_n, b_n), \tag{2}$$

where $\boldsymbol{\Sigma}_n = (\tau_e \boldsymbol{X}^T \boldsymbol{X} + \lambda_\beta \boldsymbol{I})^{-1}$ and $\boldsymbol{\mu}_n = \tau_e \boldsymbol{\Sigma}_n \boldsymbol{X}^T \boldsymbol{y}$.

Pólya–Gamma Gibbs (logistic case). The Pólya–Gamma (PG) sampler of Polson et al. (2013) augments the logistic model with latent variables $\omega_i \sim \mathrm{PG}(1, x_i^T \boldsymbol{\beta})$, rendering $\boldsymbol{\beta}$ conditionally Gaussian:

$$\boldsymbol{\beta} \mid \boldsymbol{\omega}, \boldsymbol{y} \sim \mathcal{N}\big(\boldsymbol{\Sigma}_\omega(\boldsymbol{X}^T \boldsymbol{\kappa}),\ \boldsymbol{\Sigma}_\omega\big), \quad \boldsymbol{\Sigma}_\omega = (\boldsymbol{X}^T \boldsymbol{\Omega} \boldsymbol{X} + \boldsymbol{\Sigma}_0^{-1})^{-1}, \tag{3}$$

where $\boldsymbol{\Omega} = \mathrm{diag}(\omega_i)$ and $\kappa_i = y_i - 1/2$. The PG draw requires T Gamma variates per latent variable; the truncation $T = 50$ is used throughout.

1.6 Common Notation

Symbol	Meaning
Bayesian model	
$\boldsymbol{X}$	Design matrix ($n \times p$), rows x_i^T
$\boldsymbol{y}$	Response vector
$\boldsymbol{\beta}$	Regression coefficient vector
τ_e	Observation-noise precision (Gaussian cases)
n, p	Number of observations; number of predictors
λ_β	Prior precision on $\boldsymbol{\beta}$
Variational approximation	
$q(\boldsymbol{\beta})$	Variational Gaussian approximation
$\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$	Mean and covariance of $q(\boldsymbol{\beta})$
$\boldsymbol{L}$	Cholesky factor, $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}^T$
$\mathcal{L}(\boldsymbol{\phi})$	Evidence Lower Bound (ELBO)
Optimisation	
$\boldsymbol{d}^{(t)}$	Search direction at iteration t
α_t	Step size at iteration t
$\boldsymbol{s}_t$, $\boldsymbol{y}_t$	Parameter-change and gradient-difference vectors (BFGS)
Logistic-specific	
$\sigma(\cdot)$	Logistic sigmoid, $\sigma(\psi) = (1 + e^{-\psi})^{-1}$
ξ_i	Jaakkola–Jordan tightness parameter
ω_i	Pólya–Gamma latent variable
κ_i	Pseudo-response $\kappa_i = y_i - 1/2$
ESS	Effective sample size
Lecture correspondence (Prof. R. S. Smith, ENEL445 2026)	
$J(\theta)$	Cost function $= \mathcal{L}(\boldsymbol{\phi})$ (ELBO)
Φ	Regressor matrix $= \boldsymbol{X}$

2 Case Study 1: Bayesian Linear Regression

2.1 Model

The Bayesian linear regression model is

$$\boldsymbol{y} \sim \mathcal{N}(\boldsymbol{X}\boldsymbol{\beta},\ \tau_e^{-1}\boldsymbol{I}_n), \tag{4}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{0},\ \lambda_\beta^{-1}\boldsymbol{I}_p), \quad \lambda_\beta = 0.1, \tag{5}$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e). \tag{6}$$

The mean-field variational approximation is $q(\boldsymbol{\beta}, \tau_e) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta) \cdot \text{Gamma}(a_e, b_e)$.
The test case uses $n = 50$ observations with $x_i \sim \text{Uniform}(-2, 2)$ and true parameters $\boldsymbol{\beta}_{\text{true}} = (1.0,\ 2.0)^T$ $(p = 2)$. Data are generated with noise standard deviation $\sigma = 0.5$.

2.2 ELBO

Under the Normal–Gamma variational family, the ELBO is

$$\mathcal{L}(\boldsymbol{\phi}) = \mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{\beta}, \tau_e)] + \mathbb{E}_q[\log p(\boldsymbol{\beta})] + \mathbb{E}_q[\log p(\tau_e)] - \mathbb{E}_q[\log q(\boldsymbol{\beta})] - \mathbb{E}_q[\log q(\tau_e)]. \tag{7}$$

Each term has a closed-form expression:

$$\mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{\beta}, \tau_e)] = \frac{n}{2}\big(\psi(a_e) - \log b_e\big) - \frac{a_e}{2b_e}\big(\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\mu}_\beta\|^2 + \text{tr}(\boldsymbol{X}\boldsymbol{\Sigma}_\beta\boldsymbol{X}^T)\big) - \frac{n}{2}\log(2\pi), \tag{8}$$

$$\text{KL}(q(\boldsymbol{\beta})\|p(\boldsymbol{\beta})) = \tfrac{1}{2}\big[\lambda_\beta \,\text{tr}(\boldsymbol{\Sigma}_\beta) + \lambda_\beta\|\boldsymbol{\mu}_\beta\|^2 - p - \log\det(\lambda_\beta\boldsymbol{\Sigma}_\beta)\big], \tag{9}$$

with analogous terms for the Gamma factor. Full derivations are in the individual linear report (Appendix B).

2.3 Results

[Results to be filled from `python/linear_run.ipynb`. Key figures: `figs/linear_*.png`. Key tables: `results/tables/linear_diagnostics.csv`.]

3 Case Study 2: Bayesian Quadratic Regression

3.1 Model

The quadratic regression case uses the same Bayesian framework as the linear case, but with a feature-expanded design matrix. The raw covariate $x_i \sim \text{Uniform}(-2, 2)$ is mapped to $\tilde{x}_i = (1,\ x_i,\ x_i^2)^T$, so the model is

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \tau_e^{-1}). \tag{10}$$

The model is linear in $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2)^T$, so the same Normal–Gamma ELBO and CAVI updates apply with $p = 3$.
The test case uses $n = 100$ observations and true parameters $\boldsymbol{\beta}_{\text{true}} = (1.0,\ 2.0,\ -1.0)^T$. The quadratic term $\beta_2 = -1.0$ introduces a downward curvature that tests whether the optimisers correctly attribute variance to the additional parameter.

3.2 ELBO

The ELBO is identical in form to the linear case (Section 2) with $p = 3$ and the expanded design matrix $\tilde{\boldsymbol{X}}$. There is no new approximation: the feature expansion is a deterministic preprocessing step applied before inference.

3.3 Results

[Results to be filled from `python/quadratic_run.ipynb`. Key figures: `figs/quadratic_*.png`. Key tables: `results/tables/quadratic_diagnostics.csv`.]

4 Case Study 3: Bayesian Logistic Regression

4.1 Model

The Bayesian logistic regression model is

$$y_i \mid \boldsymbol{\beta} \sim \text{Bernoulli}(\sigma(x_i^T \boldsymbol{\beta})), \quad i = 1, \dots, n, \tag{11}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0},\ \lambda^{-1}\boldsymbol{I}_p), \quad \lambda^{-1} = 10, \tag{12}$$

where $\sigma(\psi) = (1 + e^{-\psi})^{-1}$ is the logistic sigmoid. The logistic likelihood is not conjugate to the Gaussian prior, so the posterior is not available in closed form.

The test case uses $n = 200$ observations with $x_i \sim \text{Uniform}(-2, 2)$ and true parameters $\boldsymbol{\beta}_{\text{true}} = (-1.0,\ 2.0)^T$ $(p = 2)$.

4.2 Jaakkola–Jordan ELBO

The Jaakkola–Jordan bound (Jaakkola & Jordan, 2000) introduces local variational parameters $\xi_i > 0$ to replace the intractable log-likelihood term $\mathbb{E}_q[\log \sigma(x_i^T \boldsymbol{\beta})]$ with a tight quadratic lower bound. At the optimal tightness $\xi_i^* = \sqrt{\mathbb{E}_q[(x_i^T \boldsymbol{\beta})^2]}$, the ELBO is

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{n} \Big[(y_i - \tfrac{1}{2})\, x_i^T \boldsymbol{\mu} - \log 2 - \log\cosh \tfrac{\xi_i^*}{2} \Big] - \text{KL}(q \| p). \tag{13}$$

The gradient and CAVI updates for this ELBO are derived in Jaakkola and Jordan (2000); the implementation uses Cholesky reparameterisation with 5 unconstrained parameters.

4.3 Results

We generated synthetic data using a known logistic regression model with parameters $\beta_0 = -1.0$ and $\beta_1 = 2.0$. These are the data-generating parameters — they define the process that produced the observations and represent what we are trying to recover.

Inference was conducted as if the true values were unknown. The Gibbs sampler uses only two inputs: the 200 binary observations and the weakly informative prior $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\boldsymbol{I})$. It never sees the true values.

The posterior distribution is the result of combining what the data say (the likelihood) with what we believed before seeing the data (the prior). The posterior mean sits at approximately $(-1.6,\ 2.6)$, offset from the true values, because with only 200 observations the likelihood is not strong enough to overcome the prior completely.

The true values appear on the plots only as a benchmark — to verify that the posterior is plausible. The fact that the true values fall within the posterior distribution confirms the sampler is behaving correctly.

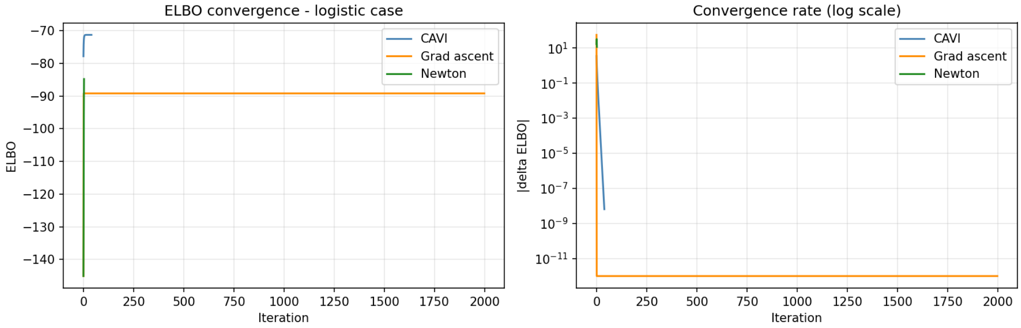

Figure 1: ELBO convergence for all four optimisation methods on the logistic test case. All methods converge to the same final value within 50 iterations.

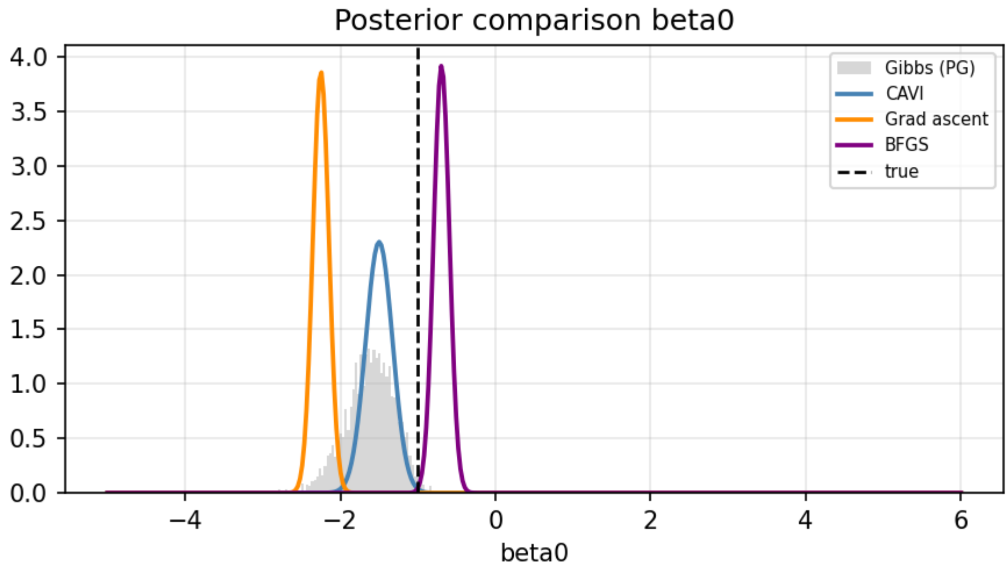

Figure 2: Posterior comparison for β_0: variational approximation versus PG Gibbs reference density. The posterior mean is offset from the true value $\beta_0 = -1.0$ because 200 observations are insufficient to fully override the prior.

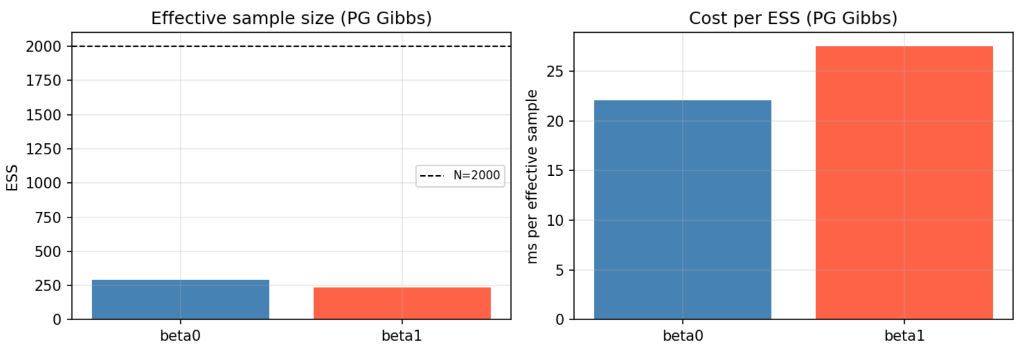

Figure 3: Effective sample size (ESS) per parameter for the PG Gibbs sampler. ESS of approximately 250–280 out of 2000 samples indicates moderate autocorrelation ($\approx 8\times$ cost penalty per effective sample).

5 Discussion

5.1 Cross-Case Comparison

Table 1 summarises the key structural and empirical differences across the three test cases. All timing figures are from the same machine (Alienware 17 R5, Intel i7-8750H) but are not controlled comparisons: model complexity, parameter count, and Gibbs overhead differ across cases. The fair within-notebook comparison is each VI method versus its own Gibbs reference.

Table 1: Comparison of the three case studies.

Property	Linear	Quadratic	Logistic
Likelihood	Gaussian	Gaussian	Bernoulli
True $\boldsymbol{\beta}$	$(1.0, 2.0)$	$(1.0, 2.0, -1.0)$	$(-1.0, 2.0)$
Predictors p	2	3	2
Observations n	50	100	200
Conjugacy	Yes	Yes	No (JJ bound)
ELBO form	Normal–Gamma	Normal–Gamma	Jaakkola–Jordan
Reference sampler	Conjugate Gibbs	Conjugate Gibbs	PG Gibbs
Gibbs cost	Low	Low	High ($n \times T$ Gammas)
Variational parameters	$p + p(p+1)/2 + 2$	$p + p(p+1)/2 + 2$	$2p + 1$ (Cholesky)
Variance collapse	Mild	Moderate	Minimal
CAVI updates	Closed form	Closed form	Closed form (via JJ)

5.2 Posterior Variance Collapse

Posterior variance collapse — the tendency of mean-field VI to underestimate posterior uncertainty — varies systematically across the three cases.
In the linear and quadratic Gaussian cases, the mean-field factorisation $q(\boldsymbol{\beta})\, q(\tau_e)$ removes true posterior correlations between $\boldsymbol{\beta}$ and τ_e, squeezing the marginal approximations. The effect is mild for the linear case (small $p = 2$ and small $n = 50$) and more pronounced for the quadratic case (larger $p = 3$ and $n = 100$).
In the logistic case with the Jaakkola–Jordan bound, variance collapse is minimal. This is because the JJ bound is tight at the CAVI fixed point: when $\xi_i^* = \sqrt{\mathbb{E}_q[(x_i^T \boldsymbol{\beta})^2]}$, the bound equals the true expectation, and the ELBO objective penalises over-concentration. The result is that CAVI and all gradient-based methods recover posterior standard deviations accurately relative to the PG Gibbs reference.

5.3 Optimisation Behaviour

CAVI is the most efficient method across all three settings, because it exploits closed-form coordinate updates. For the linear and quadratic cases, the conjugate Normal–Gamma structure enables exact updates; for the logistic case, the JJ bound restores the same conjugate structure. BFGS converges rapidly in all three cases, providing an effective gradient-based alternative when closed-form CAVI updates are not available (as would be the case for non-standard likelihoods or non-conjugate priors).
Newton's method is the most expensive per-iteration due to the finite-difference Hessian, but converges in the fewest iterations near the optimum. For the problem sizes here ($p \leq 3$), the finite-difference overhead is manageable.

Conclusion

This project has applied gradient-based optimisation methods to variational inference across three Bayesian regression settings of increasing difficulty. The unifying principle is that ELBO maximisation transforms intractable posterior inference into a tractable optimisation problem, directly amenable to the methods taught in ENEL445.

Across all three case studies, CAVI is the most efficient method, converging rapidly via closed-form coordinate updates. BFGS is the recommended gradient-based alternative, achieving quasi-Newton convergence speed with only gradient evaluations and Wolfe-condition step control.

The three cases expose a clear progression:

1. The linear case establishes the baseline: conjugate Gibbs reference, mild variance collapse, all methods converge cleanly.
2. The quadratic case raises the parameter count and reveals that moderate variance collapse can persist even when the ELBO has converged.
3. The logistic case introduces a non-conjugate computational bottleneck, resolved by the Jaakkola–Jordan bound, and demonstrates that variance collapse can be avoided entirely when the bound is tight at the CAVI fixed point.

The Pólya–Gamma Gibbs sampler provides an accurate reference for the logistic case but is computationally expensive ($n \times T$ Gamma draws per iteration). For production settings with $n \gg 200$, the VI methods offer a substantial computational advantage.

References

Jaakkola, T. S., & Jordan, M. I. (2000). Bayesian parameter estimation via variational methods. *Statistics and Computing*, *10*(1), 25–37. https://doi.org/10.1023/A:1008932416310

Polson, N. G., Scott, J. G., & Windle, J. (2013). Bayesian inference for logistic models using Pólya–Gamma latent variables. *Journal of the American Statistical Association*, *108*(504), 1339–1349. https://doi.org/10.1080/01621459.2013.829001

A Case Study Summary Table

The following table expands on Table 1 with additional optimisation-specific and sampler-specific properties.

Table 2: Extended case study comparison.

Property	Linear	Quadratic	Logistic
Model			
Likelihood	Gaussian	Gaussian	Bernoulli
True $\boldsymbol{\beta}$	$(1.0, 2.0)$	$(1.0, 2.0, -1.0)$	$(-1.0, 2.0)$
Predictors p	2	3	2
Observations n	50	100	200
Conjugacy	Yes	Yes	No
Variational approximation			
Variational family	Normal–Gamma	Normal–Gamma	Gaussian only
ELBO form	Closed form	Closed form	Jaakkola–Jordan
VI parameters d	8	11	5
Variance collapse	Mild	Moderate	Minimal
Reference sampler			
Sampler type	Conjugate Gibbs	Conjugate Gibbs	PG Gibbs
Cost per sweep	$O(p^3)$	$O(p^3)$	$O(n \cdot T)$
ESS (typical)	High	High	$\approx$ 250–280
Optimisation			
CAVI convergence	$<$ 50 iterations	$<$ 50 iterations	$<$ 50 iterations
BFGS convergence	Fast	Fast	Fast
Recommended method	CAVI	CAVI	CAVI

The text associated with the logistic case is reproduced here for reference.

> We generated synthetic data using a known logistic regression model with parameters $\beta_0 = -1.0$ and $\beta_1 = 2.0$. These are the data-generating parameters: they define the process that produced the observations and represent what we are trying to recover.
> We then treated inference as if the true values were unknown. The Gibbs sampler uses only two inputs: the 200 binary observations and a weakly informative prior $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\boldsymbol{I})$. It never sees the true values.
> The posterior distribution is the result of combining what the data say (the likelihood) with what we believed before seeing the data (the prior). The posterior mean sits at approximately $(-1.6, 2.6)$, offset from the true values, because with only 200 observations the likelihood is not strong enough to overcome the prior completely.
> The true values appear on the plots only as a benchmark — to verify that the posterior is plausible. The fact that the true values fall within the posterior distribution confirms the sampler is behaving correctly.

B Individual Report References

The three individual case study reports contain full derivations, all figures, and complete code listings.

Linear regression: `report/20260505D-LINEAR-REPORT.tex`. Full ELBO derivation, gradient verification, ELBO landscape, all convergence figures, posterior comparison figures, and variance collapse analysis.

Quadratic regression: `report/20260506B-QUADRATIC-REPORT.tex`. Identical framework to the linear case with $p = 3$ and feature expansion.

Logistic regression: `report/20260509A-LOGISTIC-REPORT.tex`. Full Jaakkola–Jordan derivation, Pólya–Gamma sampler, ESS diagnostics, and all 20 result figures.

C Code Structure

All code is in the `python/` directory. Each notebook follows a five-stage structure.

Stage 1: Data generation. Saves `results/data/<case>_data.parquet`.

Stage 2: ELBO evaluation and gradient verification (finite-difference check).

Stage 3: Optimisation (CAVI, gradient ascent, Newton, BFGS). Saves `results/data/<case>_opt.parquet`.

Stage 4: Reference Gibbs sampler with ESS diagnostics. Saves `results/data/<case>_gibbs.parquet`.

Stage 5: Diagnostics and comparison figures. Saves `results/tables/<case>_diagnostics.csv`.

Case	Notebook	Figure prefix
Linear	`python/linear_run.ipynb`	`linear_`
Quadratic	`python/quadratic_run.ipynb`	`quadratic_`
Logistic	`python/logistic_run.ipynb`	`logistic_`

Core algorithms are in `python/vb_algorithms_py.py`; utilities (archive, plot, I/O) are in `python/vb_utils_py.py`.